

Xuanzhao Gao

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RESEARCH INTERESTS

I am interested in applied and computational mathematics in general, working on the following topics:

- Boundary integral equations for complex geometries.
- Fast summation algorithms for long-range interactions.
- Tensor network-based algorithms for combinatorial optimization problems.

EDUCATION

Hong Kong University of Science and Technology, Hong Kong SAR, China 2021.9 -- 2025.7
Ph.D. in Individual Interdisciplinary Program, major in Applied Mathematics. Thesis advisors: Zecheng Gan, Jinguo Liu, and Yang Xiang

The University of Science and Technology of China, China 2017.9 -- 2021.7
B.S. in Condensed Matter Physics & B.S. in Computer Science. Undergraduate thesis advisor: Xianhui Chen

POSITIONS

Flatiron Research Fellow

Center for Computational Mathematics, Flatiron Institute, Simons Foundation, New York, NY, USA 2025.9 -- now

Visiting Scholar, collaborating with Prof. Pan Zhang

Institute of Theoretical Physics, Chinese Academy of Sciences, Beijing, China 2025.7 -- 2025.9

PUBLICATIONS

Peer-reviewed Publications

- [1] X. Gao, S. Jiang, J. Liang, and Q. Zhou, An $O(\log N)$ Monte Carlo method for periodic Coulomb systems, Accepted by SIAM J. Sci. Comput. Arxiv:2601.09288 (2026)
- [2] Z. Gan, X. Gao, and Y. Li, An $O(N)$ quasi-Ewald splitting method for nanoconfined electrostatics, Accepted by Multiscale Model. Simul. Arxiv:2601.03125 (2026)
- [3] J.-G. Liu, X. Gao, and R. Samuelson, OMEinsumContractionOrders: A Julia package for tensor network contraction order optimization, Journal of Open Source Software **11**, 9886 (2026)
- [4] X. Gao, S. Jiang, J. Liang, Z. Xu, and Q. Zhou, A fast spectral sum-of-Gaussians method for electrostatic summation in quasi-2D systems, Numerische Mathematik **158**, 533 (2026)
- [5] X. Gao, Q. Zhou, Z. Gan, and J. Liang, Accurate error estimates and optimal parameter selection in Ewald summation for dielectrically confined Coulomb systems, J. Chem. Theory Comput. **21**, 5890 (2025)
- [6] Z. Gan, X. Gao, J. Liang, and Z. Xu, Random batch Ewald method for dielectrically confined Coulomb systems, SIAM J. Sci. Comput. **47**, B846 (2025)
- [7] X. Gao, X. Li, and J. Liu, Programming guide for solving constraint satisfaction problems with tensor networks, Chinese Physics B **34**, 50201 (2025)
- [8] Z. Gan, X. Gao, J. Liang, and Z. Xu, Fast algorithm for quasi-2D Coulomb systems, J. Comput. Phys. **524**, 113733 (2025)
- [9] M. Roa-Villescas, X. Gao, S. Stuijk, H. Corporaal, and J.-G. Liu, Probabilistic inference in the era of tensor networks and differential programming, Phys. Rev. Res. **6**, 33261 (2024)
- [10] X. Gao and Z. Gan, Broken symmetries in quasi-2D charged systems via negative dielectric confinement, J. Chem. Phys. **161**, (2024)
- [11] Z. Nie, X. Gao, Y. Ren, S. Xia, Y. Wang, Y. Shi, J. Zhao, and Y. Wang, Harnessing hot phonon bottleneck in metal halide perovskite nanocrystals via interfacial electron-phonon coupling, Nano Lett. **20**, 4610 (2020)

Preprints

[12] X. Gao, L. Greengard, and S. Jiang, An adaptive fast algorithm for periodic Coulomb lattice sums in arbitrary unit cells, Arxiv Preprint Arxiv:2606.28608 (2026)

[13] Y. Wang, X. Gao, P. Zhang, F. Pan, and J. Liu, Branch-and-bound tensor networks for exact ground-state characterization, Arxiv Preprint Arxiv:2602.05470 (2026)

[14] X. Gao, Y.-J. Wang, P. Zhang, and J.-G. Liu, Automated discovery of branching rules with optimal complexity for the maximum independent set problem, Arxiv Preprint Arxiv:2412.07685 (2024)

SOFTWARE PACKAGES

PeriodicDMK: A C++ implementation of the dual-space multilevel kernel-splitting method for periodic Coulomb systems.

ExTinyMD.jl: A framework for molecular dynamics simulations.

DMK-MC: A C++ implementation of the dual-space multilevel kernel-splitting method for Monte Carlo simulation of long-range electrostatics systems.

EwaldSummations.jl: A comprehensive implementation of the Ewald summation method for electrostatic interactions in both triply and doubly periodic systems with and without dielectric mismatches.

TropicalNumbers.jl: A refined implementation of the tropical semiring.

CuTropicalGEMM.jl: A GPU-accelerated implementation of the tropical matrix multiplication.

TreeWidthSolver.jl: A collection of tools for calculating the exact tree width and tree decomposition of a given graph.

OptimalBranching.jl: A framework for automated discovery of optimal branching rules for the branch-and-bound algorithm.

OPEN SOURCE PROJECTS

Google Summer of Code 2024, The Julia Language

Contributed to the project “[Tensor network contraction order optimization and visualization](#)” released by the Julia Language community in GSoC 2024.

Open Source Promotion Plan 2023, JuliaCN

Contributed to the project “[TropicalGEMM on GPU](#)” released by the JuliaCN community in OSPP 2023.

PRESENTATIONS AND POSTERS

SciCADE 2026, Invited Talk

June 30, 2026

A Fast Algorithm for Periodic Coulomb Lattice Sums in Arbitrary Unit Cells

APS Global Physics Summit 2026, Contributed Talk

March 16, 2026

Branch-and-Bound Tensor Network for Combinatorial Optimization

The 14th CSCM Annual Conference, Invited Talk

August 17-21, 2025

A Fast Spectral Sum-of-Gaussians Method for Electrostatic Summation in Quasi-2D Systems

JuliaCN Meetup 2024, Invited Talk

Nov 2-3, 2024

TreeWidthSolver.jl: From Treewidth to Tensor Network Contraction Order

SciCADE 2024, Contributed Talk

July 15-19, 2024

Fast Algorithm for Quasi-2D Coulomb Systems

JuliaCN Meetup 2023, Contributed Talk

Dec 9, 2023

How to Implement Generic Matrix-Mul with Generic Element Types on GPU?

SKILLS

Programming Languages: Julia (proficient), Python, C/C++, CUDA

Languages: Mandarin Chinese (native), English (proficient)